NISHA SUTHAR

Machine learning | Computational Chemistry | Scientific Data Analytics

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nishasuthar | Sutharnisha

PROFESSIONAL SUMMARY

Chemistry-trained molecular ML researcher with hands-on experience in AI-driven drug discovery, molecular modelling, and scientific computing. Skilled in Python, PyTorch, RDKit, GNNs, and predictive modelling, with the ability to develop predictive models, optimize learning strategies, and tackle interdisciplinary challenges through practical, data-driven solutions. First Prize Winner at OpSciHack 2026, demonstrating innovative problem-solving and collaborative research skills.

SKILLS

Programming & Tools:	Python, R, Git, Linux, SQL, Quarto, Docker, cloud computing
Machine Learning:	Supervised learning, multi-task learning, graph neural networks, transformers.
Scientific Computing:	PyTorch, TensorFlow, Scikit-learn, RDKit, NumPy, Pandas, Matplotlib
Computational Chemistry:	QSAR modelling, Computer aided drug design, molecular-property prediction, GROMACS, VMD, ParaView
Languages:	English (C2), Hindi (Native), German (A1, currently improving)

WORK EXPERIENCE

CompagOs, Zurich, Switzerland *July 2025 - Dec 2025*
Data Science Intern - Biomedical Imaging

Tech stack: *Python, R, Git/GitHub, Quarto, Cloud computing, Software testing*

- Worked on software tools for processing and managing scientific data, including databases, testing, and GitHub.
- Refactored legacy Python code into a reproducible, cloud-based analytical system.
- Collaborated with the CEO to incorporate strategic feedback and deliver reliable analytical outputs supporting decision-making.

Teaching Assistant *July 2016 - Dec 2024*
Freelancer

- Taught advanced chemistry topic include: organic and enzyme reaction mechanisms, fragment based drug design, and structure-activity relationships in drug design.
- Mentored 70+ Bachelor's and Master's students in quantitative problem solving (linear algebra, numerical methods, QSAR)

RESEARCH PROJECTS

Multi-Task Learning for Molecular Property Prediction *March 2026 - Present*
Research Project

Tech stack: Python, PyTorch, PyTorch Geometric, Uni-Mol, GNNs, RDKit, NumPy, Pandas

- Built a reproducible PyTorch pipeline using pretrained Uni-Mol representations for multi-task prediction of quantum-mechanical molecular properties.
- Benchmarked AIM, PCGrad, and loss-scaling strategies across multiple data splits via mean-rank and relative-performance metrics.
- Analyzed task-level gradient conflicts, optimisation behaviour, and model transferability, with version-controlled, reproducible experiment configurations throughout.

Reproducible, Parameterised Life-Science Reporting Workflow *Oct 2026*
Hackathon Project – OpSciHack 2026

Tech stack: *Quarto, R/Python, Snakemake, Docker, Interactive GUI, Git/GitHub*

- Won **1st Prize** as part of a **3-member team** by developing a reproducible, parameterised workflow that generated audience-specific outputs from a shared life-science analysis pipeline.
- Automated the reporting workflow with Snakemake and Quarto to generate consistent researcher- and public-facing tables, figures, and narrative outputs.
- **Challenge Partner:** Center for Reproducible Science and Research Synthesis (CRS), University of Zurich

Machine Learning-Based Kinase Bioactivity Prediction *Feb 2025 - Jun 2025*
EPFL Course Project

Tech stack: *Python, RDKit, scikit-learn, Pandas, NumPy, Matplotlib*

- Developed a cheminformatics pipeline using ECFP fingerprints and physicochemical descriptors for kinase bioactivity prediction.
- Benchmarked SVR, Random Forest, and neural networks; the best SVR model achieved $R^2 = 0.70$ and RMSE = 0.765.

Ugi Reaction: Effect of substituents and applications *Oct 2015 - April 2016*
Master project

- Investigated the Ugi multicomponent reaction mechanism through detailed theoretical and literature-based analysis.
- Evaluated substituent effects on reaction pathways, intermediates, and potential synthetic outcomes.

EDUCATION

EPFL-Extension School, Lausanne, Switzerland *Dec 2024 - June 2025*
Certificate in Applied Data Science: Machine Learning

Kurukshetra University, Kurukshetra, India *July 2014 - June 2016*
Master of Science in Chemistry, *First Division (Distinction)*

Government National College, Sirsa, India *July 2011 - June 2014*
Bachelor of Science (Chemistry, Biology), *First Division (Distinction)*

COURSES & CERTIFICATIONS

- Supervised Machine Learning (DeepLearning.AI, Stanford University) *2024*
- Advanced Learning Algorithms (DeepLearning.AI, Stanford University) *2024*
- Unsupervised Learning, Reinforcement Learning (DeepLearning.AI, Stanford University) *2024*
- Modeling and Simulation of Biological processes-Workshop (IIT Madras, India) *2023*
- Computer Aided Drug Design (IIT Madras, India) *2023*

AWARDS & ACCOMPLISHMENTS

- 1st Prize, OpSciHack 2026 - reproducible, parameterised life-science reporting challenge by CSR *2026*
- Secured in top 0.1% in GATE-2019 exam (Graduate Aptitude Test in Engineering) *2019*
- Achieved All India Rank 37 (top 0.1%) in CSIR-NET 2018 competitive examination for lectureship eligibility *2018*
- First-place Winner: Biodiversity Poster Making Competition for innovative conservation strategies *2013*